

Startup Success Prediction

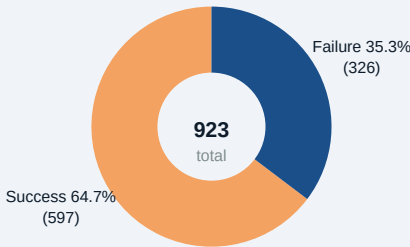
Machine Learning Project Dashboard

Dataset: 923 startups • Features: 49 • Models: Logistic Regression & Random Forest • April 2026

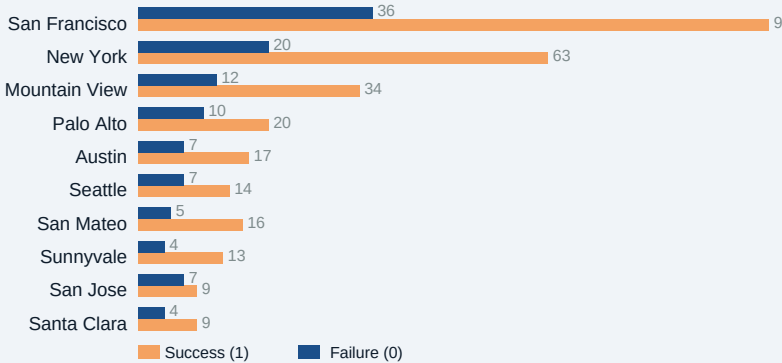


1 • DATASET OVERVIEW & CLASS DISTRIBUTION

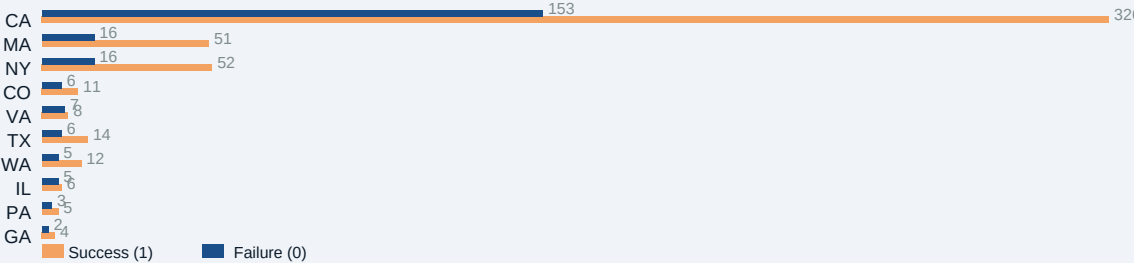
Class Distribution



Top Cities — Success vs Failure

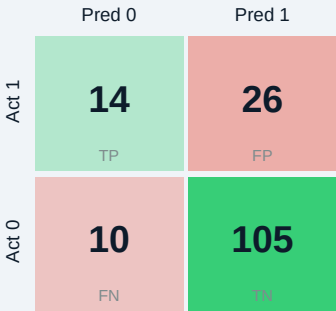


Top States — Success vs Failure



2 • MODEL PERFORMANCE

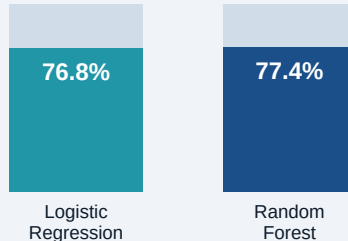
Confusion Matrix (LR)



Classification Report (LR Model)

Class	Precision	Recall	F1-Score	Support
0 (Fail)	0.58	0.35	0.44	40
1 (Success)	0.80	0.91	0.85	115
Macro Avg	0.69	0.63	0.65	155
Weighted	0.75	0.77	0.75	155

Model Accuracy Comparison



3 • FEATURE CATEGORIES USED

Category	Features	Notes
Location	latitude, longitude, state_code, is_CA, is_NY, is_MA, is_TX	Geographic indicators
Funding	funding_rounds, funding_total_usd, has_VC, has_angel, has_roundA–D	Investment signals
Temporal	age_first_funding_year, age_last_funding_year, age_first/last_milestone_year	Timing features
Network	relationships, avg_participants, is_top500	Social capital
Industry	is_software, is_web, is_mobile, is_enterprise, is_advertising, is_ecommerce, is_biotech	Sector dummies

4 • KEY INSIGHTS & RECOMMENDATIONS

KEY INSIGHTS

- California dominates with ~330 startups — 2x all other states combined; geographic concentration is a key predictor.
- 64.7% of startups in the dataset were acquired (success), indicating a moderately imbalanced dataset — SMOTE or class weights recommended.
- Random Forest (77.4%) slightly outperforms Logistic Regression (76.8%); both struggle on minority class (precision 0.58 vs 0.80).
- Features like funding rounds, has_VC, avg_participants and milestone age show strong correlation with startup success.
- Next steps: hyperparameter tuning, feature engineering on funding velocity, and ensemble methods (XGBoost/LightGBM) to push accuracy beyond 85%.